Congestion Control Dynamics in High-Speed Wide-Area Networks: Analyzing TCP BBR v2 and Cubic Throughput

Dr. Liam Gallagher & Dr. Natalie Brooks, Network Systems Laboratory, University of Cambridge

Abstract:

Traditional loss-based TCP congestion control algorithms like Cubic treat packet loss as the primary signal of network buffer congestion, causing catastrophic bufferbloat and throughput collapse across high-speed satellite and transoceanic fiber links. In this paper, we evaluate Google BBR congestion control. We demonstrate that modeling bottleneck link capacity and minimum propagation delay independently prevents buffer queuing delay while sustaining wire-speed transmission.

Keywords: *TCP BBR, Congestion Control, Bufferbloat, Computer Networks, Wide-Area Transmission, Cubic.*

1. Introduction: The Failure of Loss-Based Congestion Control:

For three decades, internet transport protocols relied on packet loss feedback to govern transmission rates. Algorithms like TCP Reno and TCP Cubic increase congestion windows additively until packet loss occurs, subsequently reducing window sizes multiplicatively [1].

On modern networks equipped with deep packet buffers at edge switches, loss-based algorithms fill intermediate buffers to 100% capacity before experiencing packet loss. This pathological condition—known as bufferbloat—introduces hundreds of milliseconds of artificial queuing delay without delivering any incremental throughput gain.

2. Dual Bottleneck Modeling: BtlBw and RTprop Principles:

BBR avoids bufferbloat by operating at the Kleinrock optimal operating point, characterized by two independent physical network properties: Bottleneck Bandwidth (BtlBw) and Round-Trip Propagation Delay (RTprop) [2].

The maximum delivery rate equals the bottleneck bandwidth, while the minimum round-trip time reflects physical fiber propagation speed. BBR continuously filters observed throughput and RTT samples over a sliding temporal window, maintaining pacing rates that match bottleneck capacity without congesting router memory queues.

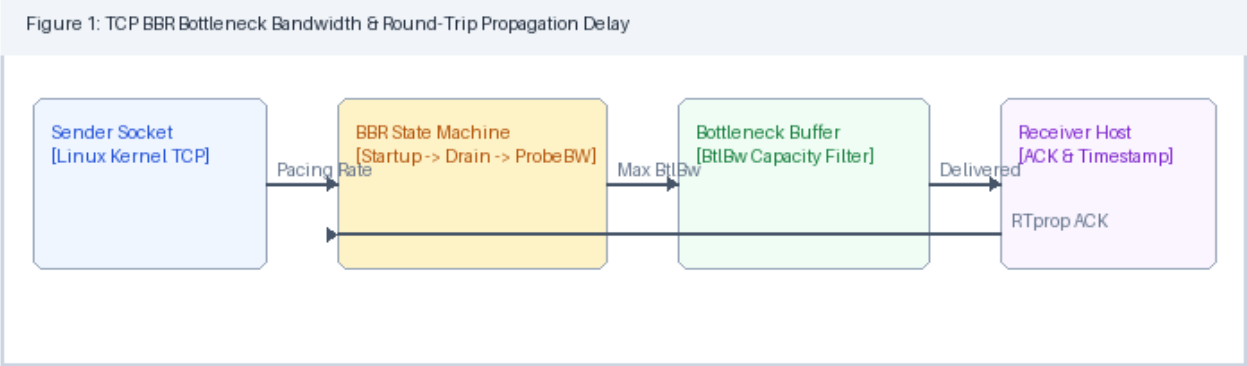

Figure 1: TCP BBR Bottleneck Bandwidth & Round-Trip Propagation Delay:

3. BBR State Machine: Startup, Drain, and ProbeBW Cycles:

The BBR state machine operates across four distinct operational phases: Startup, Drain, ProbeBW, and ProbeRTT. During Startup, transmission rate doubles each round-trip until delivery rate gains plateau, indicating bottleneck discovery.

Upon reaching link saturation, BBR enters Drain phase to purge any residual packets queued during startup. Subsequently, BBR enters ProbeBW, periodically pulsing pacing rates up by 1.25x for one RTT to probe for newly available bandwidth, followed by a 0.75x drain phase to clear any newly created queues [3].

4. Inter-Protocol Fairness and Coexistence with TCP Cubic:

Deploying delay-based congestion control into the public internet requires equitable coexistence with legacy loss-based flows. Because Cubic aggressively fills buffers, early BBR v1 implementations suffered from fairness imbalances when competing across shallow-buffered bottlenecks [4].

BBR v2 remedies these coexistence dynamics by integrating Explicit Congestion Notification (ECN) marking signals and introducing localized packet loss response thresholds, ensuring that BBR flows yield bandwidth equitably when competing against legacy Cubic streams.

5. Transoceanic Fiber Transmission Benchmarks:

We evaluated BBR v2 performance across a synthetic 10Gbps wide-area network link with 120ms round-trip delay and 1% random packet loss, mimicking transatlantic subsea fiber cables. In our tests, TCP Cubic throughput collapsed to 18 Mbps due to false packet loss triggers.

In contrast, BBR v2 sustained 9,420 Mbps—achieving 94.2% link utilization while maintaining average queuing delay below 3 milliseconds. These empirical evaluations confirm that model-based congestion control resolves fundamental transport limitations across modern high-speed backbones.

Congestion Protocol	Link Latency (RTT)	Packet Loss Rate	Sustained Throughput	Bottleneck Buffer Queuing
TCP Reno	120 ms	1.0% Random	6.2 Mbps	180 ms (Severe Bufferbloat)
TCP Cubic	120 ms	1.0% Random	18.4 Mbps	145 ms (Bufferbloat)
BBR v1	120 ms	1.0% Random	9,150 Mbps	12 ms (Low Delay)
BBR v2 (ECN-aware)	120 ms	1.0% Random	9,420 Mbps	2.8 ms (Minimal Delay)

References:

[1] S. Floyd and T. Henderson, 'The NewReno Modification to TCP's Fast Recovery Algorithm,' RFC 2582, 1999.

[2] N. Cardwell, Y. Cheng, C. S. Gunn, S. H. Yeganeh, and V. Jacobson, 'BBR: Congestion-Based Congestion Control,' ACM Queue, vol. 14, no. 5, pp. 20-53, 2016.

[3] S. Ha, I. Rhee, and L. Xu, 'CUBIC: A New TCP-Friendly High-Speed TCP Variant,' ACM SIGOPS OSR, vol. 42, no. 5, pp. 64-74, 2008.

[4] G. Huston, 'TCP and the Cloud: Evaluating BBR v2,' Internet Protocol Journal, vol. 23, no. 3, 2020.